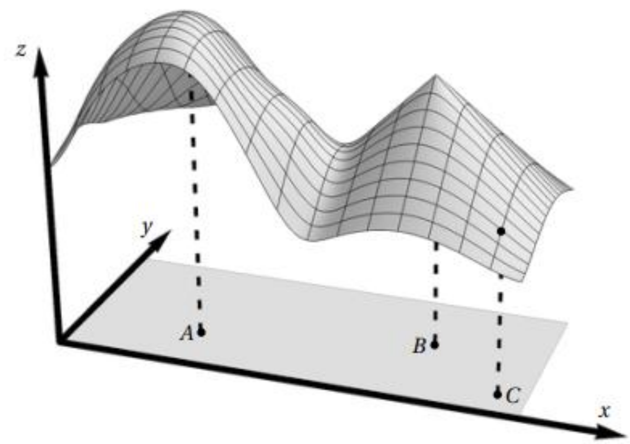
MTH 202 Chapter 3: Partial Derivatives

3.4 Differentiability Worksheet

1. Consider the function $g:\mathbb{R}^2 \to \mathbb{R}$ whose graph is shown at the right. Let A and B be points in $\mathbb{R}^2$ corresponding to two "peaks" of the graph, and C be the point in $\mathbb{R}^2$ corresponding to the dot on the graph. For each part, circle the answer that is the most consistent with the picture.



a) At the point A, the function g is:

continuous	differentiable	both	neither

b) At the point B, the function g is:

continuous	differentiable	both	neither

c) At the point C, the function $\dfrac{\partial g}{\partial x}$ is:

negative	zero	positive

2. Show that $f(x,y)=xe^{xy}$ is differentiable at $(1,0)$.

3. Show that $f(x,y)=1+x\ln(xy-5)$ is differentiable at $(2,3)$

4. Consider the function $f(x,y)=\dfrac{x^{12}+y^4+36}{3x-1}$.

a) Confirm that the function $f(x,y)$ is differentiable at the point $(1,1)$.

b) Confirm that the function $f(x,y)$ is continuous at the point $(1,1)$.

5. Consider the function $f(x,y)=\sqrt{x+e^{4y}}$.

a) Confirm that the function $f(x,y)$ is differentiable at $(3,0)$.

b) Confirm that the function $f(x,y)$ is continuous at the point $(3,0)$.